

GAN SIM RU DAYNA

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EDUCATION

National University of Singapore

Aug 2025 – Present

Bachelor of Engineering in Computer Engineering

- University Engineering Scholarship Recipient

Singapore Polytechnic

Apr 2022 – May 2025

Diploma in Mechatronics and Robotics, with Merit

- Valedictorian, Class of 2025; graduated with a GPA of 3.97/4.00
- SP Engineering Scholarship Recipient; Director's Honour Roll, AY2022–2025

EXPERIENCE

Ecovolt Technologies

Jan 2026 – Present

Hardware Engineer Intern

- Owned the hardware bring-up and customer deployment of Euna Air, an environmental monitoring and sensor-fusion solution, completing rigorous on-site validation at customer premises within one week.
- Developed embedded firmware and field-testing procedures for Euna Air, identifying discrepancies between laboratory simulations and real-world operating conditions to support reliable deployment.
- Contributed to the R&D of Maxwell Ultra, an advanced occupancy-sensing system, and streamlined its production process ahead of deployment at a hotel partner.
- Investigated complex network edge cases across enterprise IoT deployments by systematically reproducing failure modes, auditing the firmware codebase and identifying operational bottlenecks.

DSO National Laboratories

May 2026 – Jul 2026

Software Engineer Intern

- Independently delivered a three-month research project integrating R2-Dreamer, a model-based reinforcement learning algorithm, into the SMAClite R2-2100 environment for multi-agent mission-planning applications.
- Designed and executed reward-shaping, adaptive-sampling, observability and actor-critic experiments, raising validation macro win rate from near zero to a best checkpoint of 55.6% while reducing battle timeouts and remaining enemy effective hit points.
- Evaluated policy generalisation on unseen scenarios, achieving 53.11% macro win rate on the blind-IID split and 59.25% on the blind-compositional split, and identified scenario-dependent policy and value generalisation as the primary performance bottleneck.

A*STAR Institute for Infocomm Research

Sep 2024 – Feb 2025

Robotics Intern

- Spearheaded research evaluating NVIDIA Isaac Sim with ROS 2 as a potential primary simulation platform for future robotics projects at A*STAR's Institute for Infocomm Research.
- Implemented a ROSA agent using LangChain to enable natural-language control of ROS 2 robots in Isaac Sim, contributing towards more accessible interfaces for autonomous robotic systems.
- Developed realistic warehouse crowd simulations using Omni.Anim.People and conducted more than 100 simulation runs to assess Nav2 path-planning accuracy and collision-avoidance performance under dynamic human movement.

- Assembled and soldered components for three TREX robotic units, gaining hands-on experience in hardware production and prototyping.
- Designed electrical-box circuitry for the AirGuard system using Autodesk Inventor.
- Supported the on-site deployment, maintenance and troubleshooting of Camello autonomous last-mile delivery robots in Punggol.

AWARDS

- A*STAR Science Award (Polytechnic), 2023–2024
- SP Engineering Merit Award, 2023–2025
- Edusave Skills Award, 2025

LEADERSHIP AND COMMUNITY INVOLVEMENT

TOUCH Community Services

Sep 2022 – Present

Youth Mentor

- Mentor children from disadvantaged backgrounds through sustained academic support, personal guidance and enrichment activities.
- Plan and facilitate celebration and engagement programmes to foster a supportive learning environment.

SP LEO Club

Mar 2023 – Mar 2024

Welfare Head

- Organised active-ageing and outreach initiatives for senior beneficiaries, including social engagement and enrichment activities.
- Designed programmes addressing the social and emotional needs of elderly participants.

RoboCup Singapore Student Organising Committee

Oct 2022 – Apr 2023

Chairperson

- Led a student organising committee of more than 80 volunteers for the RoboCup Singapore Open, coordinating across students, mentors and external stakeholders.
- Managed event planning, volunteer scheduling and on-site operations to deliver a smooth competition experience.

School of Science and Technology

2021

President, Robotics @APEX

- Led the school robotics club and supported members in robotics training, project development and competition preparation.

TECHNICAL SKILLS

- **Programming:** Python, C++, Java, Verilog
- **Robotics and Simulation:** ROS 2 (Robotics Operating System), NVIDIA Isaac Sim, Nav2
- **Hardware and Prototyping:** CAD, Autodesk Inventor, Arduino, ESP32, STM32, Soldering, Hands-on Assembly, 3D Printing